



CLIENT WIFI NETWORK ANALYZER

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Introduction

The Client WiFi Network Analyzer is a modern application designed to monitor and analyze wireless network performance. The system provides users with insights into various network parameters such as latency, throughput, jitter, and packet loss. The project is divided into two main components: backend and frontend. The backend is responsible for authentication and database operations, while the frontend provides an interactive dashboard for visualization.

Objectives

The main objective of this project is to develop a reliable and secure system that allows users to analyze WiFi performance. It focuses on implementing secure authentication, efficient data handling, and user-friendly visualization of network statistics.

Technologies Used

The backend is developed using Node.js and Express.js, while MongoDB is used as the database. Mongoose is used for schema modeling. Additional tools include bcrypt for password hashing and JWT for authentication. The frontend is developed using HTML, CSS, JavaScript, and Chart.js for data visualization.

Backend Architecture

The backend follows a REST API architecture. It consists of server configuration, routing, and database interaction. The server is created using Express.js and listens on port 5000. Middleware such as CORS and JSON parsing are used to handle requests efficiently.

User Model

The user model is defined using Mongoose schema with fields such as username and password. Passwords are stored in encrypted format to ensure security. This model interacts directly with MongoDB.

Authentication System

The system provides two endpoints: registration and login. During registration, user credentials are received and the password is hashed using bcrypt. During login, the system verifies the credentials and generates a JSON Web Token (JWT) for secure access.

Security Features

Security is maintained through password encryption, token-based authentication, and environment variable management. These measures prevent unauthorized access and protect user data.

Frontend Architecture

The frontend is designed to provide an intuitive user interface. It consists of a login page and a dashboard page. The interface uses a clean layout and dark theme to enhance user experience.

Login System

The login functionality is implemented using JavaScript. It retrieves user input from the HTML form and validates credentials. If the credentials match predefined values, the user is redirected to the dashboard. Otherwise, an error message is displayed.

Dashboard Design

The dashboard displays key network performance indicators such as QoE score, latency, jitter, packet loss, and throughput. These values are displayed using card components for better readability and organization.

Data Visualization

Charts are created using Chart.js. A line chart is used to display latency over time, and another chart represents throughput. These visualizations help users understand network trends effectively.

Advantages

The system is secure, scalable, and easy to use. It provides real-time visualization and efficient data handling. The modular design makes it easy to extend and maintain.

Limitations

The current system uses static data for visualization and has limited validation. It does not include advanced features such as real-time monitoring or role-based access control.

Future Enhancements

Future improvements include integrating real-time WiFi monitoring, enhancing security features, adding admin controls, and improving UI/UX design.

Conclusion

The Client WiFi Network Analyzer successfully demonstrates the integration of backend and frontend technologies. It provides a strong foundation for developing advanced network analysis tools and ensures secure and efficient system performance.